TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REGISTER

Class: **Duration time: from 07/09/2026 To 30/12/2026**

(*) Profession: Software Engineer Specialty: SE

(*) Kinds of person make registers: Lecturer [x] Students

1. Register information for supervisor (if have)

No.	Fullname	Phone	**E-Mail**	**Title**
Supervisor	Tôn Thất Hoàng Minh	0936668995	minhtth5@fpt.edu.vn	Giảng viên

2. Register information for students (if have)

No.	**Full name**	**Student code**	**Phone**	**E-mail**	**Role in Group**
1	Nguyễn Thành Thắng	SE184251	0966823637	Thangntse184251@fpt.edu.vn	Leader
2	Nguyễn Đình Tuấn	SE182540	0787664860	tuanndse182540@fpt.edu.vn	Member
3	Nguyễn Phú Quý	SE180104	0766824448	Quynpse180104@fpt.edu.vn	Member
4	Phan Xuân Thịnh	SE184527	0945645753	thinhpxse184527@fpt.edu.vn	Member

3. Register content of Capstone Project

(*) 3.1. Capstone Project name:

3.1.1. English: Sales Collaborator and Affiliate Network Management System

3.1.2. Vietnamese: Hệ thống quản lý đội ngũ cộng tác viên bán hàng và tiếp thị liên kết

Project Code: FA26SE032 | Version: 2.0 (Review 2 Proposed Final Scope)

(*) 3.2. Main proposal content (including result and product)

a) Context:

Direct-to-consumer (D2C) retail conglomerates and multi-brand enterprises in Vietnam increasingly rely on external networks of sales collaborators (KOLs, KOCs, and affiliate creators) to drive product discovery and conversion across social channels. However, managing

external collaborator networks across multiple company-owned brand divisions via spreadsheets and chat groups introduces critical operational, financial, and security risks:

• **Remuneration & margin distortion:** Manual or naively stacked commission rates and unguarded price discounts erode brand unit economics, cause settlement disputes, and delay partner payouts.

• **Attribution spoofing & self-referral fraud:** Without cryptographic tracking, partner-bound coupon attribution, and self-purchase detection, enterprises suffer from bot click-farming, cookie loss, and collaborators buying through their own links to harvest rebates.

• **Reverse logistics & commission overpayment:** Paying commissions before customer return windows close causes unrecoverable cashflow loss on refunded orders, while blanket clawbacks unfairly penalize creators when carrier-damaged parcels are exchanged 1-for-1.

• **Sample sponsorship & media fragmentation:** High-resolution marketing assets are scattered across chat groups, and shipping free sample products to creators lacks structured 14-day review proof SLAs.

• **Cross-brand cannibalization & tax compliance:** Without strict category locks across internal brand divisions, database concurrency locks, and automated 10% Personal Income Tax (PIT) withholding, enterprises face internal SKU overlap and audit exposure.

Therefore, an enterprise-grade Sales Collaborator and Affiliate Network Management System (SCANMS)—delivered across Cloud-deployed Web portals and a native multi-role Mobile application—is vital to govern the end-to-end collaborator lifecycle.

b) Proposed Solutions:

SCANMS is an enterprise In-House Sales Collaborator & Affiliate Network Management System built for a multi-brand retail enterprise operating four company-owned brand divisions (Sora Skincare & Cosmetics, Aura Activewear & Apparel, GreenBio Health & Nutrition, and Lumière Smart Living), deployed across Cloud Web Portals (React 19 + Vite) and a Native Android Mobile Application (React Native 0.88 + Expo SDK 58).

1. **5-Role Enterprise IAM & Segregation of Duties (SoD):** Role-isolated access for System Administrator (exclusive internal staff & store provisioning at /admin/personnel, >16% VIP deal budget approval, executive analytics), System Manager (Maker-Checker eKYC, product & MAP pricing moderation, return arbitration, daily payout approval), In-House Shop Manager (brand catalog, P&L & collaborator operations without revenue withdrawal), Collaborator (external KOC starting at Bronze 10% tier), and authenticated Customer.

2. **Category-Locked Brand Catalog & MAP Floor-Price Governance:** 6-block 2-tier SKU variant matrix enforcing strict store Category Lock, EAN-13 barcode deduplication, Maker-Checker compliance moderation, and Minimum Advertised Price (MAP floorPrice) validation on single/batch ProductPricePromotion requests.

3. **Smart Link, Dynamic QR & KOC-Exclusive Attribution Engine:** Shortened tracking links (/r/:shortCode), SVG QR codes, 30-day AttributionSession tracking with Redis sliding-window rate limiting (100 req/min/IP), KOC-bound attribution coupons ([KOC][5|10] replacing generic retail/freeship vouchers), and automated Self-Referral Guard (forcing self-purchase commission

to 0%).
4. **Best-Benefit Precedence Remuneration, 7-Day Hold & Conditional KPI Escrow:** Non-additive order rate formula min(25%, max(VIP Deal, 10% + Tier, 10%) + Live Bonus); 7-day post-delivery pending hold for base commissions; separate conditionalDealEscrowBalance unlocking VIP deal bonuses upon KPI qualification; and closed-month clean-GMV milestone bonuses (FR-09) with automatic tier promotion.
5. **Bi-Directional Collaboration, Socket.IO VIP Deal Escalation & 14-Day Sample SLA:** Two-way store-creator cooperation, real-time Socket.IO rich deal negotiation cards (10%–16% autonomous activation; 17%–25% escalated to System Admin), GHN-tracked 0 VND sample sponsorship with a 14-day video review SLA countdown, and value-added Livestream deal pinning (+2% bonus).
6. **Dual AI Engines (4-Axis Partner Matchmaker & Fraud Sentinel):** Explainable 4-axis KOC-product scoring radar (Category synergy, Conversion Rate CR%, Tier/Social reach, AOV Price Fit) paired with anomaly detection for click flooding and suspicious referrals.
7. **Multi-Warehouse GHN Fulfillment, Dual-Path Returns & Row-Locked BIDV Payouts:** Per-brand GHN pickup warehouse dispatch; dual-path return handling where 1-for-1 EXCHANGE preserves 100% KOC commission (+ GHN carrier claim) while REFUND triggers automated commission clawback; Collaborator-only withdrawals (>= 200,000 VND) protected by PostgreSQL SELECT ... FOR UPDATE locks, flat 10% PIT double-entry FinancePosting, and BIDV Sandbox disbursement.

c) Functional Requirements:

• **Enterprise IAM, Staff Provisioning & KOC eKYC:** Authenticate 5 roles via Argon2id + JWT + 2FA OTP across Web and Mobile (/api/mobile/bootstrap); restrict internal Shop Manager/System Manager creation to System Admin; verify KOC CCCD, bank account, and social channels via Maker-Checker review.
• **Brand Product Compliance, MAP Pricing & Media Hub:** Manage category-locked 6-block SKU matrices with EAN-13 deduplication and compliance moderation; enforce SKU floorPrice on single/batch price promotions; distribute HD brand assets via a centralized Media Hub.
• **Partner Collaboration, VIP Deals, Samples & Live Pinning:** Rank KOCs via 4-axis AI matchmaking; negotiate exclusive VIP deals over Socket.IO chat with >16% Admin budget escalation; track 0 VND sample shipments and 14-day review video SLAs; support value-added live session deal pinning.
• **Multi-Channel Attribution, Checkout & Dual-Path Reverse Logistics:** Issue Smart Links, Dynamic QRs, and KOC-bound coupons ([KOC][5|10]); enforce authenticated zero-loss cart checkout (PayOS VietQR/Wallet/COD) with Self-Referral Guard; dispatch via per-brand GHN warehouses and process 7-day returns via 1-for-1 Exchange (preserving KOC fee + GHN claim) or Refund (clawback).
• **Escrow Maturation, Monthly Milestones & Tax-Compliant Payouts:** Reconcile 7-day post-delivery commission holds and KPI-gated deal escrows; settle closed-month FR-09 milestone

bonuses and tier promotions; process row-locked Collaborator withdrawals with flat 10% PIT withholding, BIDV Sandbox disbursement, and immutable AuditLog records.

d) Non-Functional Requirements:

• **Security & RBAC Isolation:** Argon2id hashing (>= 64 MB memory, 3 iterations), strict 5-role server-side guards across REST/WebSocket/Mobile bootstrap endpoints, Expo SecureStore token storage on Android, and OWASP Top-10 input validation.
• **Financial Integrity & Concurrency:** Strict PostgreSQL ACID transactions, SELECT ... FOR UPDATE row-level wallet locks preventing concurrent double-spending, double-entry FinancePosting tax accounting, and append-only AuditLog traceability.
• **Performance & Resilience:** p95 API and Smart Link redirect latency <= 200 ms; Web initial gzipped bundle <= 500 KB (FCP <= 1.5 s); Redis 100 req/min/IP sliding-window rate limiting with automatic in-memory LRU fallback on cache outage.
• **Deployability & Cross-Platform Maintainability:** Cloud-deployed NestJS Modular Monolith, React/Vite Web Frontend, PostgreSQL, and Redis staging stack, paired with a TypeScript React Native (Expo SDK 58) Android application sharing unified DTO contracts.

e) Theory & Practical:

Students apply Agile/Scrum engineering practices across full-stack web, mobile, and financial systems:
• **Modular Monolith & Realtime Backend (NestJS 11 & Socket.IO):** Domain-isolated modules, RBAC guards, @nestjs/schedule reconciliation workers, and authenticated WebSocket rooms.
• **Cross-Platform Web & Native Mobile Clients (React 19, Vite & React Native / Expo SDK 58):** Unified Warm Sand & Brand Gold design system on Web and role-scoped protected navigation on Android.
• **Relational Modeling & Concurrency Control (PostgreSQL & Prisma ORM):** 3NF schema, row-level wallet locking, state machine enforcement, and double-entry financial postings.
• **Distributed Rate Limiting & Explainable Heuristics (Redis & AI Scoring):** Sliding-window anti-spam click controls with LRU memory failover, 4-axis KOC matchmaking radar, and fraud anomaly detection.

f) Products (Expected Deliverables):

1. **Cloud-Deployed Enterprise & Brand Management Web Portal (React 19 + Vite):** Operations workspace for System Admins, System Managers, and In-House Shop Managers (staff provisioning, eKYC/product/MAP moderation, AI matchmaking, GHN fulfillment, return arbitration, BIDV payouts, and executive analytics).
2. **Collaborator & Authenticated Customer Web Experience (React 19 + Vite):** Creator marketing toolkit (Smart Links, Dynamic QRs, KOC coupons, samples, Media Hub, wallet withdrawals, live room) and authenticated customer storefront with PayOS/GHN tracking.
3. **Multi-Role Native Android Mobile Application (React Native 0.88 + Expo SDK 58):** Companion mobile app supporting all 5 roles via /api/mobile/bootstrap (catalog, KYC, realtime

chat & VIP deal negotiation, sample SLAs, MAP pricing, moderation, GHN tracking, returns, and payout oversight).
4. **Backend API & Scheduled Settlement Service (NestJS 11, Prisma, PostgreSQL & Redis):** Cloud-deployed REST + WebSocket API, escrow/SLA cron jobs, and GHN/PayOS/BIDV Sandbox adapters.
5. **Complete Engineering & Test Documentation (Reports 1–7):** SRS, SDD (ERD, State Machines, Activity Diagrams), API Specifications, CI Automated Test Suites, and Deployment Runbooks.

g) Proposed Tasks:

• **Requirement & Governance Specification (URS/SRS):** Finalize 5-role SoD boundaries, in-house 4-brand rules, Best-Benefit remuneration precedence, MAP floor-price validation, and dual-path return policies.
• **Architecture & Database Design (SDD):** Model normalized PostgreSQL schema, entity state machines (KYC, Product, Deal, Sample, Order, Return, Commission, Payout), and cross-platform Web/Mobile contracts.
• **Backend, Web & Mobile Implementation:** Build NestJS domain modules, React/Vite Web portals, and React Native/Expo Android screens with shared validation and RBAC enforcement.
• **Cloud Staging Deployment, Concurrency & E2E Verification:** Deploy Web + API + DB + Redis staging environment, build Android APK, and execute automated CI unit/integration/concurrency test suites across all 5 main workflows.

4. Other comments (propose all relative things if have).

Supervisor (If have) *(Sign and full name)* Tôn Thất Hoàng Minh	HCM, date 06/11/2026 On behalf of Registers (Sign and full name) Nguyễn Thành Thắng

Registration Form – Record of Changes

1. Project Information

Project code	FA26SE032
Project name	Sales Collaborator and Affiliate Network Management System (SCANMS)
Team	Nguyễn Thành Thắng (Leader - SE184251), Nguyễn Đình Tuấn (SE182540), Nguyễn Phú Quý (SE180104), Phan Xuân Thịnh (SE184527)
Supervisor	ThS. Tôn Thất Hoàng Minh
Current registration version	Version 1.0 (Review 1 Approved Scope)
Proposed version	Version 2.0 (Review 2 Proposed Final Scope – Enterprise In-House Affiliate + Web & Mobile)
Version	2.0
Date	06/11/2026

2. Changes

☑ Changes to the current registration form (6 architectural realignments detailed below)

No.	Section	Current content	Proposed content	Reason	Proposed by
1	**Cover & Table 2 (Duration & Team Finalization)**	Duration listed as 'from 09/2026 To 03/2027'; student table contained an unassigned 5th member slot.	Duration updated to 'from 07/09/2026 To 30/12/2026' (Fall 2026 schedule); finalized with 4 active members (Thắng, Tuấn, Quý, Thịnh).	Aligns project schedule with official Fall 2026 Capstone defense milestones and formalizes the 4-member team.	Nguyễn Thành Thắng (Leader)
2	**Section 3.2.a Context & 3.2.b Feature #2 (In-House Multi-Brand Affiliate Model & MAP Governance)**	Described open multi-merchant onboarding, flat product catalog, and unguarded retail discounts.	Realigned to an Enterprise In-House Multi-Brand Affiliate System (4 company-owned brand divisions with dedicated GHN warehouses, strict Category Lock, EAN-13 barcode deduplication, Maker-Checker product moderation, and MAP floor-price protection on ProductPricePromotion).	Refocuses scope strictly on managing external sales collaborators for an enterprise D2C conglomerate rather than building an open C2C marketplace.	Nguyễn Thành Thắng
3	**Section 3.2.b Feature #1 & 3.2.c IAM (5-Role SoD, Internal Staff Provisioning & KOC Onboarding)**	3 basic roles; allowed external shop self-registration and anonymous guest checkout.	Standardized 5 RBAC roles with Segregation of Duties (SoD): System Admin exclusively appoints internal Shop Managers & System Managers (/admin/personnel); Customers upgrade to Collaborators via Maker-Checker eKYC (starting at Bronze 10% tier); strict authenticated checkout preserves 100% cart items.	Prevents unauthorized external store creation, enforces legal KYC/bank identity matching for tax compliance, and guarantees order provenance.	Nguyễn Thành Thắng
4	**Section 3.2.b Features #3, #4 (Best-Benefit Remuneration, KOC Attribution Coupon & KPI Escrow)**	Uncapped/additive commission rules and generic retail vouchers without self-referral protection.	Adopted Best-Benefit Precedence Rate = min(25%, max(VIP Deal, 10% + Tier, 10%) + Live); KOC-Exclusive Attribution Coupons ([KOC][5|10]) replacing generic retail/freeship vouchers; Self-Referral Guard forcing self-purchase commission to 0%; separate 7-day pending hold and conditionalDealEscrowBalance for VIP KPIs.	Eliminates margin erosion from stacked discounts/commissions, blocks creator self-purchase abuse, and ensures cookie-less order attribution.	Nguyễn Đình Tuấn & Nguyễn Phú Quý

No.	Section	Current content	Proposed content	Reason	Proposed by
5	**Section 3.2.b Features #4, #7 & 3.2.c Order/Payout (GHN Fulfillment, Dual-Path Returns & BIDV Payouts)**	Linear commission settlement without return differentiation; manual bank file export without gateway verification.	Integrated per-brand GHN Sandbox fulfillment and Dual-Path Return Resolution: (a) 1-for-1 Exchange for carrier/factory damage preserving 100% KOC commission + GHN compensation claim vs (b) Refund to shopping wallet triggering commission clawback. Collaborator-only withdrawals (>= 200k VND) use SELECT ... FOR UPDATE locks, flat 10% PIT withholding, double-entry FinancePosting, and BIDV Sandbox disbursement.	Protects corporate cashflow on refunds while fairly preserving KOC earnings when damaged parcels are exchanged; enforces concurrency-safe tax accounting.	Nguyễn Phú Quý & Nguyễn Đình Tuấn
6	**Section 3.2.b Features #5, #6 & 3.2.f Deliverables (VIP Deal Escalation, Sample SLA, Dual AI, Cloud Web & Mobile App)**	Web-only static portal; no real-time deal escalation, no sample SLA tracking, no AI engines.	Added Socket.IO VIP Deal negotiation (10%–16% autonomous, 17%–25% escalated to System Admin), 14-day Sample Review SLA, 4-Axis AI Partner Matchmaking + AI Fraud Sentinel, value-added Live deal pinning, Cloud Staging/Production Web deployment, and a 5-role React Native (Expo SDK 58) Android Mobile App sharing the NestJS API.	Delivers full cross-platform mobility for creators, brand managers, and operations staff alongside cloud staging deployment required for Defense 1.1.	Phan Xuân Thịnh & Nguyễn Thành Thắng

3. Decision at Review 2 (completed by reviewers)

No.	Decision (Accepted / Adjusted / Rejected)	Reviewer note
1		
2		
3		
4		
5		
6		